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324414 (24)

BE (4th Semester)

Examination, April-May, 2013

Branch : Electrical

DIGITAL ELECTRONICS & LOGIC DESIGN

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Unit-I

Q. 1. (a) Determine the value of base "x"

$(125)_6 = (65)_x$ 2

(b) A twelve bit Hamming code was received using odd parity Check for the correctness of code 7

HC 1 0 1 0 1 0 1 0 1 0 1 1

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(c) Do as directed Write steps also.

(i) 1 9 8

+ 1 8 9 using BCD addition. 2

(ii) –5 – (–)7 = ? using 2's compliment method. 2

(iii) 496 – 794 = ? using 9's compliment. 2

(iv) $(164)_{10}$ write X – S – 3 code.

+ $(238)_{10}$ 1

(d) Define code. Discuss the various classification of code and explain each code briefly with suitable examples. 7

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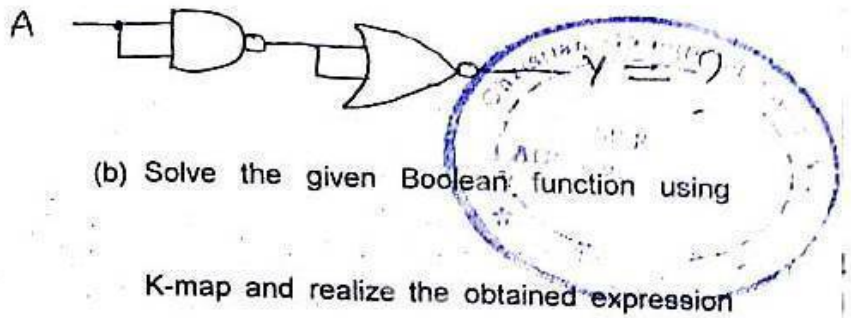
Unit-II

Q. 2. (a) Determine the output Y for given circuit 2



(b) Solve the given Boolean function using K-map and realize the obtained expression using NAND gate only 7

$Y = \Sigma m\ (1, 3, 5, 8, 9, 14, 15) + d\ (6, 7, 12, 13)$

(c) What are BASIC gates and UNIVERSAL gate. Discuss the universal property of NAND gate. 7

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(d) Minimize the given function using Quine-McClusky method and realize the circuit using BASIC gates. 7

$Y = \Sigma m\ (1, 4, 5, 10, 14) + d\ (0, 11, 15)$

Unit-III

Q. 3. (a) Write the truth table for Half Adder. 2

(b) What is PLA ? Give its basic block diagram and implement the following function using PLA. 7

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(i) $F_1 = XY + X\overline{Y}$

$F_2 = \overline{X}YZ + XZ + YZ$

(c) Implement the given Boolean function using :

(i) 16 : 1 Multiplexer 3

(ii) 8 : 1 Multiplexer 4

$Y = \Sigma m\ (1, 2, 4, 7, 9, 10, 12, 14, 15)$

(d) Design 4 bit even and odd parity generator. 7

Unit-IV

Q. 4. (a) Differentiate between latch and flip flop. 2

(b) Convert D-flip flop to T-flip flop. 7

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(c) Design Mod-5 synchronous up counter using T flip flop and also draw its timing diagram. 7

(d) What are shift registers ? Explain the basic shift operations. With the help of circuit diagram briefly discuss 4 bit PIPO registers. 7

Unit-V

Q. 5. (a) What is ROM ? 2

(b) Explain the difference between RAM and ROM. 7

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(c) Draw the basic ROM cell and explain its working.

(d) Give the comparison between Moore machine and Mealy machine used for sequential logic circuits. 7

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